Three-copy undistillability of the boundary two-ququart Werner state

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We give a computer-assisted proof that the two-ququart Werner state $\rho_* = (I - F/2)/14$ is three-copy undistillable. Equivalently, its three-factor partial-trace form is nonnegative on every complex 64×64 matrix of rank at most two, including nonnormal matrices with unequal singular values. An isometric factorization embeds every such matrix into three matched four-replica moments with valid partial-transpose and joint positivity constraints. Common symmetry averaging gives a complete rational source space of dimension 2741, including the additional alternating component present at local dimension four. An exact functional identity combines 1239 rational positive Gram multipliers with homogeneous equality constraints to reproduce the objective. Every coefficient and positivity inequality is verified in rational arithmetic, with a second implementation rebuilding the coefficient maps from literal permutation matrices. The result gives the exact three-copy distillability threshold $\alpha < -1/2$ for the two-ququart Werner family.

I. INTRODUCTION

Entanglement distillation distinguishes quantum correlations that can supply pure entangled pairs from correlations that remain inaccessible to local operations and classical communication. States with positive partial transpose are undistillable, even when arbitrarily many copies are available [1]. Whether this obstruction exhausts bound entanglement is a longstanding question. A state with negative partial transpose that remained undistillable at every copy number would establish the existence of negative-partial-transpose bound entanglement.

Werner states provide a symmetric setting in which to study this question [2, 4–6]. Their symmetry reduces the state family to one parameter, while the distillation test still ranges over arbitrary Schmidt-rank-two vectors across all available copies. The boundary of one-copy distillability is particularly useful. Proving that this boundary state remains undistillable at a larger fixed copy number isolates the possibility that collective processing may activate its entanglement. The two-ququart instance at two copies was singled out in Ref. [6], following matrix formulations developed in Ref. [7].

The finite-copy criterion admits a formulation through partial traces of low-rank coefficient matrices [8]. Recent preprints establish the sharp two-copy inequality and its Werner consequences [9–12]. Extending that argument to three copies requires new control of the simultaneous reductions of a matrix. Partial trace can increase rank, and the two-copy inequality cannot be applied successively while retaining the rank-two hypothesis. Wu and Zou prove the three-copy endpoint for all normal rank-two matrices and for specified nonnormal support classes in arbitrary local dimensions [13]. The remaining matrices need not have either of these structures.

Semidefinite formulations of Werner distillability have a substantial history [14]. A recent candidate manuscript gives a rational certificate for the unrestricted three-copy qutrit endpoint using matched four-replica moments and a coupled positive block [15]. We use that moment construction, with its unrestricted second matrix factor, and extend the source coordinates and positive representations to local dimension four. This extension introduces a fourth alternating component and its partially transposed representations. A certificate in the qutrit quotient does not control these additional directions.

Here we construct a rational certificate for the unrestricted two-ququart endpoint. The proof separates the physical reduction from a finite algebraic verification. Every complex matrix of rank at most two produces feasible matched moments. A complete rational basis represents their common symmetry average, and an exact positive-multiplier identity proves nonnegativity throughout the resulting relaxation. The certificate is accompanied by a second implementation that rebuilds its source and Gram maps before checking the rational identity. The two-copy theorems and the claimed qutrit endpoint are context for this construction, rather than premises of its proof.

II. MAIN RESULT AND THE DISTILLATION CRITERION

On $\mathbb{C}^4 \otimes \mathbb{C}^4$, the Werner family is

$$\rho(4,\alpha) = \frac{I + \alpha F}{16 + 4\alpha}, \qquad -1 \le \alpha \le 1\,, \tag{1}$$

where $F|i,j\rangle = |j,i\rangle$. The state of interest is $\rho_* = \rho(4,-1/2)$. Let $H = H_1 \otimes H_2 \otimes H_3$, with $H_i = \mathbb{C}^4$. For $C \in \mathrm{End}(H)$, partial trace contracts matching row and column indices at the indicated sites. It is unnormalized, and $\mathrm{Tr}_\varnothing C = C$. The relevant quadratic form is

$$\begin{aligned} q_3(C) &= \sum_{J\subseteq\{1,2,3\}} \left(-\frac{1}{2}\right)^{|J|} \|\mathrm{Tr}_J C\|_F^2 \\ &= \|C\|_F^2 - \frac{1}{2}\sum_{i=1}^{3} \|\mathrm{Tr}_i C\|_F^2 \\ &\quad + \frac{1}{4}\sum_{i<j} \|\mathrm{Tr}_{ij} C\|_F^2 - \frac{1}{8}|\,\mathrm{Tr}\, C|^2\,. \end{aligned} \tag{2}$$

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All matrix spaces in this paper are complex unless a real or rational coefficient space is explicitly specified. The Frobenius inner product is $\langle X, Y\rangle = \mathrm{Tr}(X^\dagger Y)$.

Theorem 1 (Three-copy endpoint)**.** *Every $C \in M_{64}(\mathbb{C})$ with* $\mathrm{rank}\, C \le 2$ *satisfies $q_3(C) \ge 0$. Consequently, $\rho_* = (I - F/2)/14$ is three-copy undistillable. The proof is computer assisted and uses the exact rational certificate specified in section V and appendix E.*

The theorem covers arbitrary input and output support planes and arbitrary singular values. The inequality is sharp. For any unit $u, v \in \mathbb{C}^4 \otimes \mathbb{C}^4$, the matrix

$$C_0 = \frac{|0\rangle\langle 0| + |1\rangle\langle 1|}{2} \otimes |u\rangle\langle v| \tag{3}$$

has rank two, $\|C_0\|_F^2 = 1/2$, and $q_3(C_0) = 0$. Indeed, the partial-trace form factorizes across this tensor product, and its first factor has one-copy value $1/2 - |1|^2/2 = 0$. For generic u, v the resulting matrix is nonnormal.

The connection to distillation follows directly from vectorization. The vector associated with C is $|\psi_C\rangle = \sum_{x,y} C_{xy}|x\rangle_A|y\rangle_B$. Its Schmidt rank across $A_1A_2A_3|B_1B_2B_3$ equals $\mathrm{rank}\, C$. With $|\Omega\rangle = \sum_{j=0}^{3} |jj\rangle$, partial transposition on Bob's factor gives

$$\rho_*^{\Gamma_B} = \frac{1}{14}\left(I - \frac{1}{2}|\Omega\rangle\langle\Omega|\right) . \tag{4}$$

Contracting the Ω factors for a subset J leaves $\mathrm{Tr}_J C$ and its complex conjugate. Expansion therefore yields

$$\langle\psi_C|\left(\rho_*^{\Gamma_B}\right)^{\otimes 3}|\psi_C\rangle = \frac{q_3(C)}{14^3} . \tag{5}$$

The Schmidt-rank-two criterion for finite-copy distillation [1, 4] proves the stated consequence. The normalized vector $\Omega/2$ has eigenvalue $-1/14$ in equation (4), confirming that ρ_* has negative partial transpose.

The endpoint also determines the full three-copy parameter threshold.

Corollary 2 (Two-ququart threshold)**.** *For $-1 \le \alpha \le 1$, the state $\rho(4, \alpha)$ is three-copy distillable if and only if $\alpha < -1/2$. In particular every state with $-1/2 \le \alpha < -1/4$ is NPT and three-copy undistillable.*

Proof. For $-1/2 \le \alpha \le 0$, a direct calculation gives

$$\rho(4, \alpha) = p\rho_* + (1 - p)\frac{I}{16}, \qquad p = \frac{-7\alpha}{4 + \alpha} \in [0, 1] . \tag{6}$$

The parties can implement this map by shared randomness, retaining their input with probability p and otherwise locally preparing maximally mixed states. Applying this local operation to each of three copies cannot turn a three-copy undistillable state into a distillable one. For $\alpha \ge 0$, the partial transpose is positive. For $\alpha < -1/2$, the vector $(|00\rangle + |11\rangle)/\sqrt{2}$ gives the one-copy expectation $(1 + 2\alpha)/(16 + 4\alpha) < 0$. Tensoring it with $|0\rangle_A|1\rangle_B$ on each remaining copy preserves Schmidt rank two and gives a negative three-copy expectation. Finally, the only potentially negative eigenvalue of $\rho(4, \alpha)^{\Gamma_B}$ is $(1 + 4\alpha)/(16 + 4\alpha)$, which proves the NPT statement. □

A. Relation to the fourth and fifth problems

Problem 4 of Ref. [6] asks whether an NPT state can remain undistillable at every finite copy number. For a fixed state ρ, the distillation criterion is

$$\rho \text{ is distillable} \iff \exists n \ge 1,\ \exists \psi, \quad \mathrm{SR}(\psi) \le 2, \quad \langle\psi|(\rho^{\Gamma_B})^{\otimes n}|\psi\rangle < 0 . \tag{7}$$

Here Schmidt rank refers to the bipartition between all of Alice's systems and all of Bob's systems. The two-dimensional local projections can act jointly on all copies held by each party. Their output is a two-qubit NPT state, which is distillable [1, 4]. For NPT bound entanglement, the same state must fail this test for every n. Establishing failure at one fixed n does not supply that quantifier.

The existence question can be restricted to Werner states without loss of generality [3, 4, 7]. For the nontrivial direction, suppose that ρ is NPT and undistillable. A vector ψ has negative expectation against ρ^{Γ_B}. After embedding the local spaces in equal dimension, a local filter L can be chosen with $(L^\dagger \otimes I)\Omega_d$ proportional to ψ. Its successful output σ satisfies $\mathrm{Tr}(F\sigma) = \langle\Omega_d|\sigma^{\Gamma_B}|\Omega_d\rangle < 0$. Averaging σ under $U \otimes U$ produces a Werner state and preserves this negative swap expectation, hence its NPT property. Filtering and averaging are local operations with classical communication. A distillation protocol for their output would also distill the input, which proves the required preservation of undistillability. The converse follows because Werner states are bipartite states.

In the convention $\rho(d, \alpha) = (I + \alpha F)/(d^2 + \alpha d)$, NPT is equivalent to $\alpha < -1/d$. One-copy distillability is equivalent to $\alpha < -1/2$. Indeed, the squared overlap of a normalized Schmidt-rank-two vector with the unnormalized $\Omega_d = \sum_{j=0}^{d-1} |jj\rangle$ is at most two, and that value is attained. Problem 4 is therefore equivalent to the existence of $d > 2$ and $\alpha \in [-1/2, -1/d)$ such that

$$\langle\psi|[\rho(d, \alpha)^{\Gamma_B}]^{\otimes n}|\psi\rangle \ge 0 \tag{8}$$

for every $n \ge 1$ and every ψ of Schmidt rank at most two. The dimension and the state are chosen once, before the universal quantifier over the copy number.

Problem 5 selects $d = 4$, $\alpha = -1/2$, and $n = 2$ [6]. Theorem 1 concerns the same state at $n = 3$. Selecting this endpoint is a sufficient route to a positive solution of Problem 4 if it works for every n. It is not an equivalent restriction of its existence question. Failure of this particular route would not exclude other NPT bound entangled Werner states.

The distinguished dimension four makes the partial transpose proportional to a reflection. With $P = |\Omega\rangle\langle\Omega|/4$, its unnormalized form is $I - 2P$. For two copies,

$$14^2(\rho_*^{\Gamma_B})^{\otimes 2} = I - 2Q_2, \qquad Q_2 = P \otimes (I - P) + (I - P) \otimes P . \tag{9}$$

Two-copy undistillability is equivalent to the half-property $\langle\psi|Q_2|\psi\rangle \le 1/2$ for every normalized Schmidt-rank-two vector [7]. Under vectorization, the normalized vectors in $\mathrm{ran}\, Q_2$ have coefficient matrices $K = A \otimes I_4 + I_4 \otimes B$, where

$\operatorname{Tr} A = \operatorname{Tr} B = 0$ and $\|A\|_F^2 + \|B\|_F^2 = 1/4$. Optimizing the squared overlap of a fixed vector with Schmidt-rank-two vectors gives the sum of its two largest squared Schmidt coefficients. Exchanging the two maximizations consequently gives the equivalent matrix inequality

$$s_1(K)^2 + s_2(K)^2 \le \frac{1}{2} . \tag{10}$$

These are singular values of K, or square roots of eigenvalues of $K^\dagger K$. Eigenvalues of K alone do not suffice for nonnormal matrices. This is the reduction in Eqs. (50)–(52) of Ref. [7].

For any copy number, vectorization provides a second exact matrix formulation. If $C \in M_{4^n}(\mathbb{C})$, the same contraction argument as in equation (5) gives

$$\begin{aligned} q_n(C) &= \sum_{J \subseteq \{1,\dots,n\}} \left(-\frac{1}{2}\right)^{|J|} \|\operatorname{Tr}_J C\|_F^2 , \\ \langle \psi_C | (\rho_*^{\Gamma_B})^{\otimes n} | \psi_C \rangle &= 14^{-n} q_n(C) . \end{aligned} \tag{11}$$

The rank of C equals the Schmidt rank of ψ_C. Thus nonnegativity of q_n on all complex matrices of rank at most two is equivalent to n-copy undistillability of ρ_*. No Hermiticity, normality, or equality of singular values is assumed. The case $n = 2$ gives Problem 5, and $n = 3$ gives the present theorem. The matched moments below provide an outer relaxation of this exact matrix problem. Their positivity gives a sufficient certificate and does not assert that every feasible moment has a physical realization.

We now establish theorem 1. The proof has three components. An exact lifting produces valid positive moments for every matrix in the theorem. Complete source coordinates represent their symmetry average. A rational certificate expresses the objective as a sum of positive pairings modulo valid equalities.

III. MATCHED MOMENTS FOR ARBITRARY COMPLEX RANK-TWO MATRICES

A. An isometric lifting

Homogeneity allows a nonzero C to be normalized by $\|C\|_F^2 = 1/2$. An isometry $A \in M_{64,2}(\mathbb{C})$ with $\operatorname{ran} C \subseteq \operatorname{ran} A$ and the matrix $B = 2C^\dagger A$ then satisfy

$$C = \frac{1}{2} A B^\dagger, \qquad A^\dagger A = I_2, \qquad \|B\|_F^2 = 2 . \tag{12}$$

The first identity follows from $AA^\dagger C = C$. For the last one, $\|B\|_F^2 = 4 \operatorname{Tr}(C^\dagger A A^\dagger C) = 4 \|C\|_F^2$. A rank-one range can always be completed to the required two-dimensional range. There is no condition $B^\dagger B = I_2$. In particular the lifting preserves unequal singular values of C.

The unit vectors $a = \operatorname{vec}(A)/\sqrt{2}$ and $b = \operatorname{vec}(B)/\sqrt{2}$ lie in $\mathcal{V} = H \otimes Q$, where $Q = \mathbb{C}^2$ and $\dim \mathcal{V} = 128$. The reduced state of a on Q is $I_2/2$. For two replicas of $\mathcal{V}$, let F_i swap their H_i factors and let F_Q swap their Q factors. The swap identity takes the form

$$\langle a \otimes b | F_Q \prod_{j \in J} F_j | a \otimes b \rangle = \|\operatorname{Tr}_J C\|_F^2 \qquad (J \subseteq \{1,2,3\}) . \tag{13}$$

To check the conjugations, let s, t index the traced sites and u, v the remaining sites. Both sides equal

$$\frac{1}{4} \sum_{s,t,u,v,\mu,\nu} A_{su,\mu} \overline{B_{sv,\mu} A_{tu,\nu}} B_{tv,\nu} . \tag{14}$$

It follows that

$$q_3(C) = \langle a \otimes b | F_Q \prod_{i=1}^{3} \left(I - \frac{1}{2} F_i\right) | a \otimes b \rangle . \tag{15}$$

B. Moment equalities and positive maps

The matched-moment construction uses four replicas of $\mathcal{V}$. These replicas are auxiliary moments and are distinct from the three resource copies in the distillation problem. For $P_v = |v\rangle\langle v|$, the three moments are

$$\begin{aligned} Y_{31} &= P_a \otimes P_a \otimes P_a \otimes P_b , \\ Y_{22} &= P_a \otimes P_a \otimes P_b \otimes P_b , \\ Y_{13} &= P_b \otimes P_b \otimes P_b \otimes P_a . \end{aligned} \tag{16}$$

The order in Y_{13} is part of the convention. Each moment and every partial transpose on complete replicas are positive semidefinite. This follows because each is a tensor product of positive rank-one projectors, and transposition on a complete replica replaces its vector by its complex conjugate.

The normalization of a, b gives two matching marginals. Deleting replica 3 from Y_{31} and Y_{22} leaves $P_a \otimes P_a \otimes P_b$. Deleting replica 2 from Y_{22} leaves $P_a \otimes P_b \otimes P_b$. Deleting replica 2 from Y_{13} and ordering its remaining replicas as $(4, 1, 3)$ gives the same operator. For a replica r occupied by a, the conditional equality is

$$\operatorname{Tr}_{H_r} Y_f = \frac{I_{Q_r}}{2} \otimes \operatorname{Tr}_{H_r Q_r} Y_f . \tag{17}$$

The tensor factors on the right are restored to their original positions. The construction imposes this equality at $r = 1$ for $f = 31, 22$ and at $r = 4$ for $f = 13$. The identical-factor symmetries imply the analogous equalities at the other a replicas. No such condition is imposed at a b replica.

A coupled positive block retains information shared by the three moments [15]. The vectors $z = a \otimes a \otimes b \otimes \bar{a}$ and $w = b \otimes b \otimes a \otimes \bar{b}$ give

$$\mathcal{J}(Y) = \begin{pmatrix} zz^\dagger & zw^\dagger \\ wz^\dagger & ww^\dagger \end{pmatrix} \succeq 0 . \tag{18}$$

All four entries are linear images of the moments. The convention $V_p |x_1, x_2, x_3, x_4\rangle = |x_{p^{-1}(1)}, x_{p^{-1}(2)}, x_{p^{-1}(3)}, x_{p^{-1}(4)}\rangle$

fixes the permutations $\tau = (3\,4)$ and $\pi = (1\,3\,2\,4)$. The linear maps are

$$\mathcal{D}(Y) = (V_\tau Y V_\tau^{\mathsf{T}})^{\Gamma_4}, \qquad \mathcal{K}(Y) = (Y V_\pi)^{\Gamma_4}\,, \tag{19}$$

and the block is

$$\mathcal{J}(Y) = \begin{pmatrix} \mathcal{D}(Y_{31}) & \mathcal{K}(Y_{22}) \\ \mathcal{K}(Y_{22})^\dagger & \mathcal{D}(Y_{13}) \end{pmatrix}\,. \tag{20}$$

For example, with arbitrary row and column multi-indices x, y in $\mathcal{V}^{\otimes 4}$, the upper right entry has matrix element

$$a_{x_1} a_{x_2} b_{x_3} \bar{a}_{x_4} \bar{b}_{y_1} \bar{b}_{y_2} \bar{a}_{y_3} b_{y_4} = z_x \bar{w}_y\,. \tag{21}$$

This follows by right multiplication by V_π followed by transposition on replica 4. The diagonal identities follow by exchanging replicas 3 and 4 before that transposition. Thus the block inequality holds for general complex a, b.

C. Common averaging

We apply the same averaging operations to all three moments. For $U = U_1 \otimes U_2 \otimes U_3 \otimes U_Q$, with $U_i \in U(4)$ and $U_Q \in U(2)$, average Y_f under $U^{\otimes 4}$. Also average over permutations of the three physical sites and over complex conjugation. The resulting moments are denoted $\overline{Y}_f$. Every equality above is preserved. For a complete-replica transpose, unitary conjugation becomes conjugation by the corresponding tensor product of U and $\bar{U}$, which preserves positivity.

The coupled block is preserved as a whole. Under $a \mapsto Ua$ and $b \mapsto Ub$, both z and w transform by $U^{\otimes 3} \otimes \bar{U}$. Physical-site permutations likewise act identically on the two components. Complex conjugation sends the whole block to its complex conjugate. The average is therefore positive. Its entries remain the same linear images of the averaged moments. This common averaging is necessary to preserve the cross entry.

The objective is unchanged. It is real and invariant under all these operations. In the four-replica notation it is $\mathrm{Tr}(O Y_{31})$, where

$$O = F_Q^{(1,4)} \prod_{i=1}^{3} \left(I - \frac{1}{2} F_i^{(1,4)} \right)\,. \tag{22}$$

The two unused replicas have unit trace. Consequently every original matrix gives averaged moments with objective $q_3(C)$, valid homogeneous equalities, and valid positive blocks. This is an outer relaxation of the physical moment set. No converse realization of arbitrary feasible moments is needed. In particular, using real symmetric averaged moments imposes no reality condition on C.

IV. COMPLETE RATIONAL COORDINATES AND POSITIVE COMPRESSIONS

A. Source generators and their normalization

Regroup the four replicas by physical site and auxiliary qubit. Schur–Weyl duality places the local unitary commutant in the span of

$$D_{\boldsymbol{p}} = V_{p_1}^{(4)} \otimes V_{p_2}^{(4)} \otimes V_{p_3}^{(4)} \otimes V_{p_Q}^{(2)}, \qquad \boldsymbol{p} \in \mathfrak{S}_4^4\,. \tag{23}$$

The 24 permutations at $d = 4$ are independent. The auxiliary permutation span at $d = 2$ has dimension 14.

Let $R_{31} = R_{13}$ be the subgroup of $\mathfrak{S}_4$ fixing replica 4, and let R_{22} preserve the two sets $\{1, 2\}$ and $\{3, 4\}$. The global identical-factor projector is $\Pi_f = |R_f|^{-1} \sum_{r \in R_f} V_r^{(\mathcal{V})}$. The moment $\overline{Y}_f$ is supported on its range. For W_σ permuting the physical sites simultaneously in all replicas, the source projection is

$$\mathcal{P}_f(X) = \frac{1}{12} \sum_{\sigma \in \mathfrak{S}_3} W_\sigma \left[\Pi_f X \Pi_f + (\Pi_f X \Pi_f)^{\mathsf{T}} \right] W_\sigma^{\mathsf{T}}\,. \tag{24}$$

It is an idempotent real linear map on the local commutant. The source generators are $B_{f,k} = \mathcal{P}_f(D_{\boldsymbol{p}_{f,k}})$. The chosen words $\boldsymbol{p}_{f,k}$ are recorded in the certificate.

Lemma 3 (Complete source spaces)**.** *The selected generators form rational bases of the source spaces for $\overline{Y}_{31}, \overline{Y}_{22}, \overline{Y}_{13}$. Their dimensions are* 667, 1407, 667*, respectively.*

Proof. The symmetry properties place each averaged moment in $\mathrm{ran}\,\mathcal{P}_f$. The trace of this idempotent is its rank. The exact trace calculation in appendix B gives 667 for $f = 31, 13$ and 1407 for $f = 22$. Reducing the selected generators in a literal local permutation basis produces integer coefficient matrices after columnwise denominator clearing. Their square minors have nonzero determinants 686863, 137174, and 686863 modulo 1000033. These determinants prove independence over $\mathbb{Q}$ and $\mathbb{R}$. The independence lower bounds equal the source dimension upper bounds, which proves spanning. □

The stored coordinate convention includes a positive scale $s = 128^4 = 268435456$. For $x = (x_{31}, x_{22}, x_{13}) \in \mathbb{R}^{2741}$, define $\mathcal{Y}_f(x) = \sum_k x_{f,k} B_{f,k}$. The coordinates of physical moments obey

$$\mathcal{Y}_f(x) = s \overline{Y}_f\,. \tag{25}$$

The stored objective has $c_{31,k} = s^{-1} \mathrm{Tr}(O B_{31,k})$ and $c_{22,k} = c_{13,k} = 0$. It follows that $c^{\mathsf{T}} x = q_3(C)$ exactly. Equality and positive maps act on $\mathcal{Y}_f(x)$. Their common factor s preserves the homogeneous equalities and positive semidefiniteness.

Expanding the conditional and matching maps in independent output coordinates gives a 2912×2741 integer matrix E after a common denominator 1152 is cleared. Every physical coordinate vector satisfies $Ex = 0$. Its exact rational rank is 272. The certificate uses the submatrix E_0 comprising the 272 selected rows listed in its equality file. Their rank is also 272. Explicit partial-trace and ordering conventions are given in appendix A.

At dimension four, the source space includes an additional local alternating component. The antisymmetrizer $\mathcal{A}_4^{(d)} = (1/24) \sum_{p \in \mathfrak{S}_4} \mathrm{sgn}(p) V_p^{(d)}$ is zero at $d = 3$ and has rank one at $d = 4$. Its marginal is

$$\mathrm{Tr}_r\, \mathcal{A}_4^{(d)} = \frac{d-3}{4} \mathcal{A}_3^{(d)}\,. \tag{26}$$

Support on the alternating subspace and unitary invariance imply proportionality, and traces give the factor $\binom{d}{4}/\binom{d}{3}$. The dimension-four construction retains this direction and its partially transposed representations. In particular the source calculation uses all 24 physical permutations rather than the 23-dimensional qutrit quotient.

B. Literal representations and Gram maps

For each used local dimension d and replica cut T, the archive supplies rational embeddings J_λ and local matrices $\rho_\lambda(p)$ satisfying the literal identities

$$(V_p^{(d)})^{\Gamma_T} J_\lambda = J_\lambda \rho_\lambda(p), \qquad G_\lambda = J_\lambda^{\mathsf{T}} J_\lambda \succ 0 \,. \tag{27}$$

Here J_λ has full column rank. All 24 identities are verified for each of the 50 local embeddings. Tensoring four such embeddings gives L_t and $G_t = L_t^{\mathsf{T}} L_t$ for each parent block. For a source block obtained from $\mathcal{Y}_f(x)^{\Gamma_T}$, the rational matrix $\mathscr{R}_t(x)$ then obeys

$$L_t^{\mathsf{T}} \mathcal{Y}_f(x)^{\Gamma_T} L_t = G_t \mathscr{R}_t(x) \,. \tag{28}$$

The right side is symmetric, even when $\mathscr{R}_t(x)$ is not symmetric in its nonorthonormal coordinates.

For each retained block the archive gives a rational frame N_t. Its positive map is

$$\widehat{M}_t(x) = N_t^{\mathsf{T}} G_t \mathscr{R}_t(x) N_t \,. \tag{29}$$

This is a congruence of a positive physical moment block, multiplied by s. For joint blocks, the same construction uses $\mathrm{diag}(L_t, L_t)$ in equation (20) and a single joint frame N_t. The off-diagonal entry is retained before compression. The explicit metric-adjoint convention is stated in appendix C.

No completeness property of the frames is needed for this positivity implication. Any rational frame defines a valid positive compression. Their selection and the equality families used to find them concern the discovery of a certificate, not the coverage of the original matrices. Completeness of the source coordinates in lemma 3 ensures that a functional identity checked there holds for every physical averaged moment.

V. A RATIONAL GRAM CERTIFICATE FOR THE THREE-COPY ENDPOINT

The finite certificate consists of rational symmetric matrices Z_t, rational congruences R_t, and a rational equality multiplier η. The frames, embeddings, source words, and equality row labels specify the linear maps appearing with these multipliers. The counts relevant to the proof are collected in table I.

Proposition 4 (Finite rational certificate)**.** *The supplied data satisfy the functional identity*

$$c^{\mathsf{T}} x = \sum_{t=1}^{1239} \mathrm{Tr}\big[Z_t \widehat{M}_t(x)\big] + \eta^{\mathsf{T}} E_0 x \qquad \text{for every } x \in \mathbb{R}^{2741} \,. \tag{30}$$

Certificate quantity	Value
Source dimensions $(31, 22, 13)$	$(667, 1407, 667)$
Total source coordinates	2741
Equality matrix size	2912×2741
Selected equality rows	272
Positive Gram multipliers	1239
Multipliers on joint blocks	111
Largest Gram order	43
Independent symmetric Gram entries	18753
Rows in the positivity checks	4231

TABLE I. Exact certificate dimensions. These counts describe the supplied construction and are not asserted to be minimal.

For each multiplier, $Z_t = Z_t^{\mathsf{T}} \in M_{k_t}(\mathbb{Q})$ and its saved rational congruence satisfies, for $1 \le i \le k_t$,

$$(R_t Z_t R_t^{\mathsf{T}})_{ii} - \sum_{j \neq i} |(R_t Z_t R_t^{\mathsf{T}})_{ij}| > \frac{9999992}{10000000} \,. \tag{31}$$

In particular every Z_t is positive definite.

Exact verification. The source generators, literal module matrices, and frames determine rational coefficient maps. They are rebuilt using integer contractions with explicit bounds excluding overflow, as described in appendix C. For each standard coordinate vector $e_k \in \mathbb{R}^{2741}$, the verifier computes the right side of equation (30) over $\mathbb{Q}$ and checks equality to c_k. All 2741 differences are exactly zero. Linearity proves the identity for arbitrary real coordinates.

The verifier reads the rational entries of Z_t and R_t, checks their dimensions and symmetry, and evaluates the left side of equation (31) over $\mathbb{Q}$. All 4231 inequalities satisfy the stated rational lower bound. A real symmetric matrix with positive diagonal-dominance margins is positive definite. Indeed, bounding each cross term in its quadratic form by $2|v_i v_j| \le v_i^2 + v_j^2$ gives a strictly positive lower bound for every nonzero vector. Thus $R_t Z_t R_t^{\mathsf{T}} \succ 0$. This forces R_t to be nonsingular, and congruence implies $Z_t \succ 0$. The saved inputs, the verification procedure, and the resulting records are specified in appendix E. □

Proof of theorem 1. For $C \neq 0$, first normalize $\|C\|_F^2 = 1/2$. The lifting and common averaging in section III produce physical moments. By lemma 3, they have source coordinates x satisfying $Ex = 0$ and $c^{\mathsf{T}} x = q_3(C)$. Every $\widehat{M}_t(x)$ is positive semidefinite by equations (28) and (29), with the same statement for the joint compressions. Proposition 4 therefore gives

$$q_3(C) = \sum_{t=1}^{1239} \mathrm{Tr}\Big[Z_t^{1/2} \widehat{M}_t(x) Z_t^{1/2}\Big] \ge 0 \,. \tag{32}$$

Homogeneity restores an arbitrary norm, and $C = 0$ is immediate. Equation (5) and the finite-copy distillation criterion complete the proof. □

The positivity of the stored multipliers is strict in their compressed coordinates. The resulting physical inequality is not

strict because some physical moment compressions vanish, including those arising from equation (3). The proof uses no floating-point eigenvalue test, numerical rank decision, or optimization stopping tolerance. Numerical optimization was used to discover the data, after which an exact affine correction and exact positive-congruence checks produced the certificate. Details of that distinction are given in appendix D.

VI. DISCUSSION

The three-copy endpoint at local dimension four is controlled by an exact positive identity on matched moments of arbitrary rank-two coefficient matrices. The joint block relates the three moment families through off-diagonal entries, while the conditional auxiliary-qubit marginals encode the isometric support of the original matrix. The dimension-four alternating component is retained throughout. These ingredients permit a certificate that applies without structural assumptions on the input matrix.

The result establishes the full three-copy threshold for the two-ququart Werner family. It also applies to smaller local coefficient spaces that embed isometrically in each $\mathbb{C}^4$ factor, since such embeddings preserve all the partial-trace norms in equation (2). For the NPT interval $-1/2 \le \alpha < -1/4$, collective processing of three copies does not enable distillation. The proof does not control four or arbitrarily many resource copies. NPT bound entanglement requires undistillability at every finite copy number and is therefore not established here.

Extending the certificate to larger copy numbers requires joint control of the positive compressions and the matched-moment equalities. Whether these constraints admit a positive recursion remains open.

Appendix A: Permutation conventions and homogeneous equalities

1. Diagram conventions

All permutation lists in the data are lexicographically ordered. A stored permutation is a tuple $(p(0), p(1), p(2), p(3))$, with labels starting at zero. A stored word is a tuple of four indices into this 24-element list, in the order of three physical sites followed by Q. The permutation matrices obey $V_p V_q = V_{p\circ q}$ and $V_p^{\mathsf{T}} = V_{p^{-1}}$. The maps in equation (19) therefore replace each local word entry according to

$$p \longmapsto \tau p \tau^{-1} \quad \text{or} \quad p \longmapsto p\pi \tag{A1}$$

before taking the indicated partial transpose. In the stored labels, $\tau = (0, 1, 3, 2)$ and $\pi = (2, 3, 1, 0)$. For a partially transposed diagram, permutation multiplication is performed before transposition. The matrices representing partially transposed diagrams are not assumed to obey the original permutation group multiplication law.

The real local permutation span is denoted $\mathfrak{C}_d$. For $d = 4$, all 24 diagrams are independent because their actions on $|0, 1, 2, 3\rangle$ give distinct basis vectors. For $d = 2$, an independent list of permutation indices is

$$\mathcal{I}_2 = (0, 3, 4, 7, 8, 9, 11, 12, 13, 15, 16, 18, 20, 23)\ . \tag{A2}$$

Its Gram matrix is nonsingular, and all 24 literal matrices lie in its span. Both facts are checked over $\mathbb{Q}$. If $Q_{a,p}$ denotes the coordinate of $V_p^{(2)}$ in this list, then

$$V_p^{(2)} = \sum_{a=1}^{14} Q_{a,p} V^{(2)}_{\mathcal{I}_2(a)}\ . \tag{A3}$$

The second implementation obtains Q directly from the 16×16 permutation matrices and verifies the reconstructed matrices entrywise.

2. Matching and conditional maps

The equality maps can be specified before any coordinate reduction. Let $\mathcal{T}_r$ trace the entire replica $\mathcal{V}_r$ and retain the other replicas in increasing order. Let $\mathcal{R}$ on $\mathcal{T}_2 Y_{13}$ put its original labels in the order $(4, 1, 3)$. The two matching equations are

$$\begin{aligned} \mathcal{T}_3 Y_{31} - \mathcal{T}_3 Y_{22} &= 0, \\ \mathcal{T}_2 Y_{22} - \mathcal{R}\mathcal{T}_2 Y_{13}\mathcal{R}^{\mathsf{T}} &= 0\ . \end{aligned} \tag{A4}$$

For a selected a replica, the conditional map is

$$\mathcal{S}_r(Y) = \mathrm{Tr}_{H_r}\, Y - \iota_r\left(\frac{I_{Q_r}}{2} \otimes \mathcal{T}_r Y\right)\ , \tag{A5}$$

where ι_r restores the original positions of the remaining factors. The three conditional equations are $\mathcal{S}_1(Y_{31}) = 0$, $\mathcal{S}_1(Y_{22}) = 0$, and $\mathcal{S}_4(Y_{13}) = 0$.

After tracing the physical factors of one replica, each remaining physical site has three replicas. Its six permutation diagrams are independent at $d = 4$. Averaging over physical-site permutations leaves $\binom{6+3-1}{3} = 56$ orbit coordinates. The untraced four-replica auxiliary algebra has dimension 14, giving $56 \cdot 14 = 784$ output coordinates for each conditional map. In a matching map the auxiliary system also has three replicas. Its permutation span has dimension five, because the three-fold antisymmetrizer vanishes at $d = 2$. Thus each matching map has $56 \cdot 5 = 280$ coordinates. The total output size is $3 \cdot 784 + 2 \cdot 280 = 2912$.

The stored row order consists of the three conditional maps for $(31, 22, 13)$, followed by the aab and abb matching maps. Within each output, physical permutation triples are in increasing lexicographic order after sorting, followed by the auxiliary index. Coefficients refer to orbit averages, rather than orbit sums. Expanding the five maps on each source basis generator and clearing the common denominator 1152 gives the integer matrix E. The selected row labels in the certificate also start at zero. The multiplier η is paired with these integer rows, so no additional factor 1152 belongs in equation (30).

The model uses these homogeneous equalities. Although physical moments have unit trace, an affine trace constraint

is not used in the certificate identity. Multiplication of all moments by the positive constant s in equation (25) is therefore compatible with every equality in the proof. The source and equality data are reconstructed from literal partial traces in the independent verifier, including all orderings in equation (A4).

Appendix B: Completeness of the source basis

1. An exact projection-rank calculation

Before applying the identical-factor projectors, the physical-site invariant part of the commutant is $\mathrm{Sym}^3(\mathfrak{C}_4)\otimes\mathfrak{C}_2$. The three operations in equation (24) commute after averaging over the identical-factor subgroups. The transpose operation commutes with the sandwich projector because Π_f is real symmetric. Physical-site permutations commute with simultaneous replica permutations. Consequently $\mathcal{P}_f$ is idempotent on the relevant commutant.

For $h,k\in R_f$ and $\epsilon\in\{0,1\}$, the local linear map is

$$K^{(d)}_{h,k,\epsilon}(X)=V^{(d)}_h X^{\mathsf{T}\epsilon} V^{(d)}_k\,, \tag{B1}$$

where $X^{\mathsf{T}0}=X$ and $X^{\mathsf{T}1}=X^{\mathsf{T}}$. Its traces on the two local permutation spans determine the global projection trace. With $t_j=\mathrm{Tr}[(K^{(4)}_{h,k,\epsilon})^j]$ and $u=\mathrm{Tr}[K^{(2)}_{h,k,\epsilon}]$, the symmetric-cube identity is

$$\mathrm{Tr}\big[\mathrm{Sym}^3(K^{(4)}_{h,k,\epsilon})\big]=\frac{t_1^3+3t_1t_2+2t_3}{6}\,. \tag{B2}$$

To derive it, insert the average of the six factor permutations into $K^{\otimes 3}$. The identity permutation contributes t_1^3, the three transpositions contribute t_1t_2, and the two three-cycles contribute t_3. Therefore

$$\dim\mathrm{ran}\,\mathcal{P}_f=\frac{1}{2|R_f|^2}\sum_{h,k\in R_f}\sum_{\epsilon=0}^{1}\frac{t_1^3+3t_1t_2+2t_3}{6}\,u\,. \tag{B3}$$

This calculation needs only literal permutation actions and the auxiliary quotient in equation (A3). At $d=4$, K sends V_p to V_{hpk} when $\epsilon=0$ and to $V_{hp^{-1}k}$ when $\epsilon=1$. Its power traces count fixed labels in the 24-element basis. At $d=2$, the same images are reduced in the 14-element basis before taking the trace. The exact sums are given in table II.

Family	$\epsilon=0$ sum	$\epsilon=1$ sum	$2\lvert R_f\rvert^2$	Rank
31	37584	10440	72	667
22	37984	7040	32	1407
13	37584	10440	72	667

TABLE II. The two sums include the factor $1/6$ from equation (B2). Adding them and dividing by $2|R_f|^2$ gives the exact source dimension.

2. Independent selected generators

For each word in the saved basis list, the verifier reconstructs $\mathcal{P}_f(D_{\boldsymbol{p}})$ by enumerating both sandwich subgroup elements, all six physical-site permutations, and the transpose choice. The common diagram denominator is $12|R_f|^2$, hence 432 for 31, 13 and 192 for 22. The resulting full diagram coefficient matrix is compared exactly with the stored source table.

For the independence check, the auxiliary diagrams are reduced through equation (A3) and identical physical orbits are collected. If an orbit coefficient is collected as a total weight, its basis element is the corresponding orbit average. This convention fixes the factors for orbits containing repeated permutations. After clearing denominators in each column, the saved pivot rows select a square integer matrix. Its determinant is evaluated modulo 1000033. The nonzero values quoted in lemma 3 prove that the integer determinants are nonzero. Only this lower-bound implication is used from modular arithmetic. The matching upper bounds come from the rational idempotent trace, not from a numerical or modular rank estimate alone.

Finally, the objective can be reconstructed independently of any moment sample. For each generator, $c_{31,k}=s^{-1}\,\mathrm{Tr}(OB_{31,k})$ is evaluated using literal local matrix traces, while the other two families have zero objective coefficients. For a local permutation, $\mathrm{Tr}\,V^{(d)}_p=d^{\#\mathrm{cycles}(p)}$. Insertion of a swap replaces p by its product with that transposition. These exact traces verify the normalization as well as the sign of every term in equation (2).

Appendix C: Local embeddings and reconstruction of the Gram maps

1. Physical meaning of the saved representations

The retained source cuts, stated with replica labels starting at one, are listed in table III. Every cut transposes an entire replica of $\mathcal{V}$, including its auxiliary qubit. Joint blocks use the single transpose at replica 4 that occurs in equation (19). A transpose at only a physical subfactor would not have the positivity justification used in this proof.

Family	Complete-replica cuts
31, 13	$\varnothing,\{1\},\{4\},\{1,2\}$
22	$\varnothing,\{1\},\{3\},\{1,2\},\{1,3\}$
Joint	$\{4\}$ after the word maps in equation (A1)

TABLE III. Positive maps used in the certificate. Selecting a subset of valid positive maps does not restrict the original physical inputs.

For each cut type, the local module dimensions are

$$d=4\quad\begin{cases}1,1,2,3,3 & |T|=0,1,\\ 1,1,1,1,2,4 & |T|=2,\end{cases}$$
$$d=2\quad 1,2,3\qquad(|T|=0,1,2)\,. \tag{C1}$$

The embeddings for the different positions of a cut are included explicitly. Their squared dimensions sum to 24 at $d=4$ and 14

at $d = 2$. The positivity argument uses the literal full-column-rank embeddings, so it does not require a classification or completeness assertion for the local module list.

All entries of J_λ, G_λ, and $\rho_\lambda(p)$ are rational. The verifier constructs $(V_p^{(d)})^{\Gamma_T}$ from the identity tensor by permuting output indices and then exchanging the row and column indices at each transposed replica. It checks equation (27) and full column rank of J_λ. Thus $G_\lambda \succ 0$ follows from a literal Gram matrix, without an eigenvalue tolerance.

In nonorthonormal coordinates the adjoint relation is

$$G_t \mathscr{R}_t(X^\dagger) = \mathscr{R}_t(X)^\dagger G_t \ . \tag{C2}$$

For real averaged source coordinates, let $A_t(x), C_t(x), D_t(x)$ be the represented top diagonal, cross, and bottom diagonal entries of a joint block. Before the saved joint frame is applied, the symmetric compression is

$$H_t(x) = \begin{pmatrix} G_t A_t(x) & G_t C_t(x) \\ C_t(x)^\mathsf{T} G_t & G_t D_t(x) \end{pmatrix} \ . \tag{C3}$$

It equals a congruence of the literal physical joint block by $\mathrm{diag}(L_t, L_t)$. The positive map used in the certificate is $\widehat{M}_t(x) = N_t^\mathsf{T} H_t(x) N_t$. In particular the lower cross entry is the transpose of $G_t C_t(x)$, not a bare transpose of a nonorthonormal representation followed by an unjustified metric placement.

Before the final frames, the selected construction has 1722 parent blocks, of which 702 involve an additional dimension-four local module. The retained maps comprise 335 source blocks for each of 31, 13, 458 for 22, and 111 joint blocks. Different frames may be associated with one parent block. Of the 1239 retained maps, 522 involve an additional dimension-four local module. The archive gives each parent, cut, local module tuple, frame, and order explicitly. These labels are checked when a multiplier is read.

2. Rational coefficient reconstruction

For a Gram matrix of order k_t, the stored symmetric coordinates are Z_{ij} with $j = 1, \dots, k_t$ and $i = j, \dots, k_t$, in that order. Let $E^{[ij]}$ be E_{ii} for $i = j$ and $E_{ij} + E_{ji}$ otherwise. The coefficient map is determined by

$$(L_t)_{k,[ij]} = \mathrm{Tr}\Big[E^{[ij]} \widehat{M}_t(e_k)\Big] \ . \tag{C4}$$

For an off-diagonal coordinate this includes both symmetric entries. No $\sqrt{2}$ factor is used in these exact stored coordinates.

An explicit reconstruction uses $W = N_t E^{[ij]} N_t^\mathsf{T}$. For a source block it contracts W against the tensor products of the local matrices $G_\lambda \rho_\lambda(p)$, followed by the rational source diagram table. For a joint block, partition W into its two-by-two blocks. The pairing with equation (C3) is

$$\mathrm{Tr}(W H_t) = \mathrm{Tr}(W_{11} G_t A_t) + 2\,\mathrm{Tr}(W_{21} G_t C_t) + \mathrm{Tr}(W_{22} G_t D_t) \ . \tag{C5}$$

This accounts for the factor two in every joint cross contribution. The word maps in equation (A1) are applied locally before the partially transposed representations are selected.

All input rational matrices are first expressed as integer matrices with common positive denominators. Tensor contractions then use integer arithmetic. Before a product AB, an integer bound

$$|(AB)_{ij}| \le \left(\sum_\ell |A_{i\ell}|\right) \max_{\ell,j} |B_{\ell j}| \tag{C6}$$

controls every intermediate sum as well as the final entry. The second implementation checks bounds with arbitrary-precision Python integers before using fixed-width integer products. All accepted bounds are below 2^{62}. Sparse source contractions are bounded in the same way. Denominators are kept separately and common factors are canceled exactly. Thus the independently reconstructed coefficient maps contain no coefficients inferred by rounding a floating-point result.

The final identity check evaluates

$$r_k = c_k - \sum_t \mathrm{Tr}\Big[Z_t \widehat{M}_t(e_k)\Big] - \eta^\mathsf{T} E_0 e_k \tag{C7}$$

for all $k = 1, \dots, 2741$ in rational arithmetic. Every r_k is exactly zero. There is no omitted-coordinate or least-squares tolerance in this check. Together with lemma 3, this verifies the identity on all averaged physical inputs.

Appendix D: Discovery of the certificate and exact reconstruction

1. Equality states and multiplier ranges

Known zero states help identify useful multiplier ranges in a degenerate semidefinite problem. If an identity with positive multipliers holds at a physical zero x_*, then each term $\mathrm{Tr}[Z_t \widehat{M}_t(x_*)]$ is nonnegative and their sum is zero. It follows that

$$\mathrm{ran}\, Z_t \subseteq \ker \widehat{M}_t(x_*) \ . \tag{D1}$$

Indeed, zero trace of a product of positive matrices gives $\widehat{M}_t(x_*)^{1/2} Z_t^{1/2} = 0$. Intersecting such kernel constraints implements the relevant part of facial reduction [16]. The product zeros in equation (3) already impose restrictions, but additional nonnormal zero families constrain further cross entries.

A useful family lies on embedded qubit supports. Let $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $\Theta = J^{\otimes 3}\mathsf{K}$, where K is complex conjugation. The antiunitary obeys $\Theta^2 = -I$. For unit $u, v \in (\mathbb{C}^2)^{\otimes 3}$ satisfying $\langle u, \Theta v\rangle = 0$, define

$$C_{u,v} = \frac{1}{2}\big(|u\rangle\langle v| + |\Theta v\rangle\langle \Theta u|\big) \ . \tag{D2}$$

The two pairs $(u, \Theta v)$ and $(v, \Theta u)$ are orthonormal. Thus $C_{u,v}$ has rank two and squared Frobenius norm $1/2$. Its zero objective follows from the linear involution $\mathcal{S}(C) = J^{\otimes 3} C^\mathsf{T} (J^{\otimes 3})^\dagger$.

This involution exchanges the two summands and acts by $(-1)^w$ on a Pauli string of weight w. The weight-three coefficients of $C_{u,v}$ therefore vanish.

To translate that fact into q_3, the maps $\mathcal{D}_i(C) = C - (I_{2,i}/2) \otimes \mathrm{Tr}_i\, C$ are commuting orthogonal projections on the embedded qubit matrix space. Their product selects precisely weight-three Pauli strings, and expansion gives

$$q_3(C) = \langle C, \mathcal{D}_1\mathcal{D}_2\mathcal{D}_3(C)\rangle = \|\mathcal{D}_1\mathcal{D}_2\mathcal{D}_3(C)\|_F^2 \ . \qquad \text{(D3)}$$

Hence $q_3(C_{u,v}) = 0$. Local isometric embedding in $\mathbb{C}^4$ preserves every partial-trace norm, so the same zero is valid for the target dimension. The balanced choice $v = u$ gives the anti-state family discussed in Ref. [13]. Allowing distinct u, v also gives nonnormal members.

For example, a real seed and its paired vector are

$$u = \frac{1}{5}(1, 2, -1, 3, 2, -2, 1, 1)^{\mathsf{T}}, \qquad v = (\sigma_z \otimes \sigma_z \otimes I_2)u \ . \qquad \text{(D4)}$$

They satisfy the required orthogonality. For $C_{u,v} = C_0/50$, the sums of squared partial-trace norms of the integer matrix C_0 at subset sizes $0, 1, 2, 3$ are $(1250, 2742, 726, 484)$. The identity $8(1250) - 4(2742) + 2(726) - 484 = 0$ gives its exact zero. It is nonnormal because

$$\left\|C_{u,v}C_{u,v}^{\dagger} - C_{u,v}^{\dagger}C_{u,v}\right\|_F^2 = \frac{1260000}{50^4} > 0 \ . \qquad \text{(D5)}$$

These families were used to select the frames supplied with the certificate. Rational averaging of parameter families can also use unitary permutation integration [17]. Their role is to guide the search. The final proof needs only that each saved frame is a valid congruence and that the complete functional identity and multiplier positivity hold.

2. From a numerical candidate to a rational identity

The numerical search optimized complete Gram matrices in the retained frames. A candidate with a prescribed positive margin in its numerical coordinates was converted to rational Gram factors by dyadic rounding. Factor products give initially positive semidefinite rational matrices. They do not automatically satisfy the desired coefficient identity.

An exact affine correction was then computed together with the equality multipliers. After clearing denominators in the coefficient map, a chosen 2634×2634 integer minor was solved over $\mathbb{Q}$. The resulting correction was checked against all 2741 original source coordinates, including the 107 coordinates outside that minor. Thus a successful solve of a selected subsystem was not treated as a proof of the full identity.

An affine correction can destroy positivity. After the correction, a separate verification therefore tested every final Gram matrix by its saved rational congruence. The lower bound in equation (31) applies to these final matrices. Its value is a margin for the transformed Gram matrices, not a positive lower bound for $q_3(C)$, which has exact zeros. The final certificate can be checked without rerunning any optimization, choosing a numerical rank threshold, or reproducing this correction.

Appendix E: Certificate data and reproducible verification

1. Data needed to reproduce the proof

The companion repository[1] contains the manuscript source, the exact data, and the independent verifier. It preserves the relative directory `research/matched_d4`. Within that directory, `source.json` gives the selected words, normalization, objective, and source-basis information. The three `diagrams_*.npz` files store integer source diagram tables with their denominators specified in the source file. The file `equality_integer.npz` contains E. Local embeddings and diagram images are in `aligned_modules.json`, and parent block labels and replica cuts are in `positive_blocks.json`.

The subdirectory `full_gram_crossed` contains `exact_frames.json` and `frames.json`, which specify the rational frames and their block labels. Its `rational_maps` directory contains the rational Gram coefficient maps, represented as integer matrices with scalar denominators. The `rational_certificate` directory contains files `0000.json` through `1238.json`, each recording a multiplier Z_t, its congruence R_t, and its labels. The file `equality.json` records the selected integer row indices and the rational entries of η. Rational entries are serialized as integer or fraction strings.

The original input manifest records 2485 hashes covering the multipliers and their coordinate inputs. The independent verifier checks this pre-existing manifest without replacing it. The source diagram tables are additionally reconstructed from their literal defining words. The companion repository includes a separate package manifest for the manuscript, verifier, and all distributed files. Hashes identify an exact set of inputs. Their agreement is an integrity check and is not substituted for any algebraic verification.

The dimension-four source-coordinate construction used permutation conventions and auxiliary quotient data from the MIT-licensed code accompanying Ref. [15]. The dimension-four literal embeddings, additional source directions, frames, and present certificate were constructed in this project. The independent audit program imports no construction program or earlier certificate verifier. It uses literal permutations, the serialized rational inputs, and general-purpose arithmetic libraries. The companion repository retains the upstream attribution and license information for the reused material.

2. Verification stages

The independent program is `research/d4_independent_certificate_audit.py`. The five stages can be run from the repository root.

```
cd research
audit=d4_independent_certificate_audit.py
python "$audit" reduction
```

[1] https://github.com/GLiuax/d4-three-copy-werner

```
python "$audit" foundation
python "$audit" rank
python "$audit" maps --workers 3
python "$audit" certificate
```

The stages perform the following checks.

1. The reduction stage compares formal matrix-entry monomials for all three joint identities. It also performs numerical diagnostics of the swap and vectorization conventions on general complex inputs, including rank one and unequal singular values. The numerical diagnostics are not premises of the universal proof, which is the index argument in section III.

2. The foundation stage checks the original input hashes, all 50 literal module embeddings and 1200 intertwining equations, the source projection dimensions, and nonzero basis minors. It reconstructs every source generator and every equality and objective column using literal partial traces and matrix traces.

3. The rank stage verifies over $\mathbb{Q}$ that both E and the selected row submatrix have rank 272. For the positivity implication, validity of the selected rows alone would suffice.

4. The maps stage rebuilds all 1239 rational Gram maps using the integer contractions described in appendix C. It checks exact agreement with the saved maps and writes the independently rebuilt maps to a separate audit directory.

5. The certificate stage uses the rebuilt maps and objective to check all 2741 rational identities. It checks symmetry and the strict rational diagonal-dominance inequalities for all 1239 multipliers, covering 4231 rows in total.

The recorded rational run used NumPy 2.4.4, SciPy 1.17.1, SymPy 1.14.0, and python-flint 0.9.0. The coefficient identity and final positivity checks use exact rational arithmetic. Integer operations used for the map reconstruction are bounded as in equation (C6). The earlier verifier additionally checked positivity by rigorous Arb interval arithmetic at 256-bit precision, but the rational checks alone suffice for the certificate.

All reported checks passed on 14 October 2026. The records are stored in `research/matched_d4/full_gram_crossed/audit_2026-10-14`. The two implementations were developed within the same project. Their agreement supplies implementation diversity and does not constitute external peer review or formal verification in a proof assistant. The mathematical reduction, source-space completeness, and the finite rational checks together constitute the computer-assisted proof.

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